

MIT OCW Computational
structures
7, 14, 15

Chapter 7 : Pipelined Performance

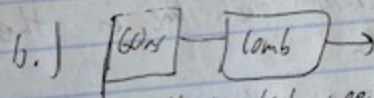
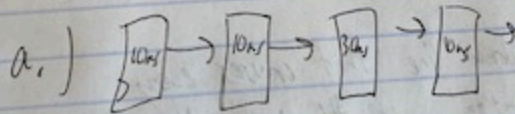
Latency is a measure of time to complete a single task.

Throughput is a measure of the number of tasks that can be completed at once.

Pipeline stages

More pipeline stages increases throughput, but decreases latency, not ideal for smaller loads, or real time loads.

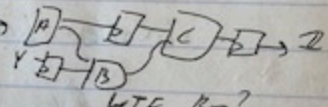
Ex.



If individual tasks use broken into smaller ones to be solved in sequential steps (pipelining), example one would have a throughput of 2 output per 30ns. Example two not as high.

Even though option 1 has higher throughput, it takes an input 120 ns to complete vs option 2's 60 ns. This demonstrates the tradeoff.

□ Balancing a pipeline could mitigate this, as a pipeline is only as good as its longest delay, but no matter what, latency is always lower due to FF delays that would otherwise not be present.

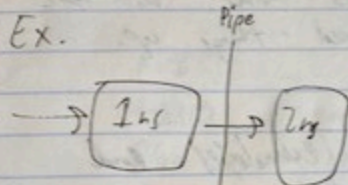
Successive Inputs should NEVER get mixed. Ex. → 

Pipelining Method

- 1 Draw line that crosses every output in the circuit. Make sure every connection crosses each line in the same direction
- 1 Place registers around the longest combinational components.

Goal, achieve max throughput with fewest registers.

What if one stage cannot be broken up and I still want to speed up Pipeline?

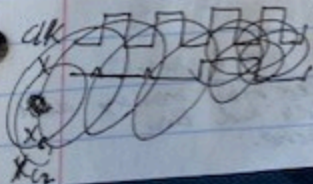
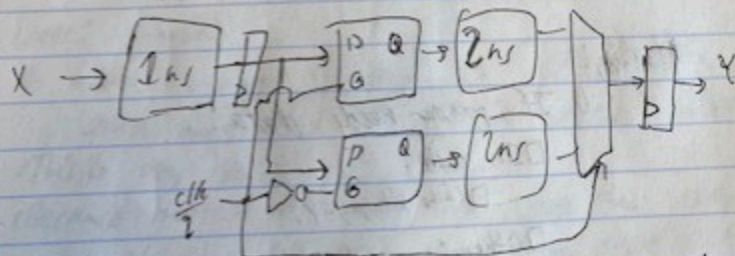
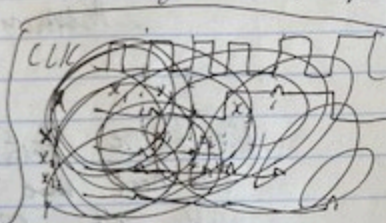


Latency: 3ns
throughput: $\frac{1}{3}$ /ns

not pipelined
Latency: 3ns
throughput: $\frac{1}{3}$ /ns

Better idea

Now try this



Still takes 3ns to get original signal, but having parallel duplicates doubles the speed of that stage. ~~Latency~~ throughput now 2/ns.

Handshaking protocols involving
wait/ready signals can ~~really~~ ~~prevent~~
help prevent deadlocks or loops
if conditionals are used correctly.
This is much less stringent than
strictly-timed pipelines. Making them
asynchronous ~~or~~ even purely
combinational is another hot topic
that could speed things up.

Memory Technologies and Control

Registers are fast, but not very spacious, so
we need more distant memory to work
with like SRAM and DRAM.

Caching:

0 If processor needs data:

□ In cache:

□ Cache hit, yay!

□ Otherwise, cache miss

↳ Minimize these please, now
we have to access main memory and
find the address and then
store on cache.